

Project 5: Cryptography MATH 4175

Participating Students:

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Questions 1-2:

Selected Plaintext is 49 20 6E 65 65 64 20 61 6E 20 41 2B 20 70 6C 7A

Selected Key0 is 45 6E 63 72 79 70 74 20 6D 79 20 6D 61 72 6B 73

Selected Key1 is 04 11 EC 9D 7D 61 98 BD 10 18 B8 D0 71 6A D3 A3

Plaintext Matrix Notation:

```
+-----+-----+-----+-----+
| 49 | 65 | 6E | 20 |
+-----+-----+-----+-----+
| 20 | 64 | 20 | 70 |
+-----+-----+-----+-----+
| 6E | 20 | 41 | 6C |
+-----+-----+-----+-----+
| 65 | 61 | 2B | 7A |
+-----+-----+-----+-----+
```

Key (K0) Matrix Notation:

```
+-----+-----+-----+-----+
| 45 | 79 | 6D | 61 |
+-----+-----+-----+-----+
| 6E | 70 | 79 | 72 |
+-----+-----+-----+-----+
| 63 | 74 | 20 | 6B |
+-----+-----+-----+-----+
| 72 | 20 | 6D | 73 |
+-----+-----+-----+-----+
```

Key (K1) Matrix Notation:

```
+-----+-----+-----+-----+
| 04 | 7D | 10 | 71 |
+-----+-----+-----+-----+
| 11 | 61 | 18 | 6A |
+-----+-----+-----+-----+
| EC | 98 | B8 | D3 |
+-----+-----+-----+-----+
| 9D | BD | D0 | A3 |
+-----+-----+-----+-----+
```

Question 3:

3a. After XOR (AddRoundKey) K0:

```
+-----+-----+-----+-----+
| 0C | 1C | 03 | 41 |
+-----+-----+-----+-----+
| 4E | 14 | 59 | 02 |
+-----+-----+-----+-----+
| 0D | 54 | 61 | 07 |
+-----+-----+-----+-----+
| 17 | 41 | 46 | 09 |
+-----+-----+-----+-----+
```

3b. After SubBytes:

```
+-----+-----+-----+-----+
| FE | 9C | 7B | 83 |
+-----+-----+-----+-----+
| 2F | FA | CB | 77 |
+-----+-----+-----+-----+
```

	D7		20		EF		C5	
+	-----	+	-----	+	-----	+	-----	+
	F0		83		5A		01	
+	-----	+	-----	+	-----	+	-----	+

3c. After ShiftRows:

+	-----	+	-----	+	-----	+	-----	+
	FE		9C		7B		83	
+	-----	+	-----	+	-----	+	-----	+
	FA		CB		77		2F	
+	-----	+	-----	+	-----	+	-----	+
	EF		C5		D7		20	
+	-----	+	-----	+	-----	+	-----	+
	01		F0		83		5A	
+	-----	+	-----	+	-----	+	-----	+

3d. After MixColumns:

+	-----	+	-----	+	-----	+	-----	+
	1C		50		3B		16	
+	-----	+	-----	+	-----	+	-----	+
	3A		B5		74		E7	
+	-----	+	-----	+	-----	+	-----	+
	C2		CD		27		02	
+	-----	+	-----	+	-----	+	-----	+
	0E		4A		30		25	
+	-----	+	-----	+	-----	+	-----	+

3e. After XOR (AddRoundKey) K1:

+	-----	+	-----	+	-----	+	-----	+
	18		2D		2B		67	
+	-----	+	-----	+	-----	+	-----	+
	2B		D4		6C		8D	
+	-----	+	-----	+	-----	+	-----	+
	2E		55		9F		D1	
+	-----	+	-----	+	-----	+	-----	+
	93		F7		E0		86	
+	-----	+	-----	+	-----	+	-----	+